Animal Behavior Observation Lab (Webcam Ethogram + Chi-Square)

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Learning goals

By the end of this lab you should be able to:

- Use an **ethogram** to record animal behavior with **operational definitions**.
- Collect data using **scan sampling**.
- Write a clear **null and alternative hypothesis** about behavior.
- Use a **chi-square test** to evaluate your hypothesis.

Part 1 (in lab): Mystery box

(Complete the mystery box activity as instructed.)

Part 2: Observing animal behavior (webcam)

Materials

- Live animal webcam/video
- This handout
- Timer (phone timer is fine)
- Data table (below)

Video Feed Choices

San Diego Zoo https://zoo.sandiegozoo.org/live-cameras

Lincoln Park Zoo https://www.lpzoo.org/live-cams/

Smithsonian National Zoo https://nationalzoo.si.edu/webcams

Explore.org Live Feeds https://explore.org/livecams/currently-live

Important observation rules

- Score only what you can **see**, not what you think the animal "means."
- If the animal is partly hidden and you **cannot determine the behavior**, use:
  - **Behavior Obscured** (you know where it is, behavior unclear)
  - **Animal Not Visible** (you can't see it at all)

Step A — Choose a focal animal and a "comparison"

You will test a question using **two conditions**. Choose ONE comparison:

Option 1 (recommended): Time comparison

- Condition A: first 10 minutes of observation
- Condition B: second 10 minutes of observation

Option 2: Two animals

- Animal A vs Animal B (only if both are visible reliably)

Option 3: Two exhibit zones

- Zone A vs Zone B (only if the exhibit is clearly divided and you can tell zones)

Write your choice here:

Comparison type: ______________________________

Condition A: ______________________________

Condition B: ______________________________

Step B — Pick 5 behavior categories (simplify the ethogram)

The full ethogram is extensive. For this lab, you will collapse behaviors into **5 simple categories** so your chi-square test stays clean.

Use these categories (circle the exact codes you will use):

1. **Inactive** (Inactive / Standing / Sitting / Lying / Floating / Vertical clinging / Hanging)
2. **Feeding/Foraging/Drinking** (Feeding/Foraging/Drinking + all foraging subtypes + Feeding/Drinking/Ruminating/Nursing)
3. **Locomotion** (Walking/Running/Climbing/Leaping/Swimming/Flying/etc.)
4. **Other (Solitary)** (Exploration, self-maintenance, scent marking, solitary play, elimination, vocalization, etc.)
5. **Social** (Affiliative + Reproductive + Agonistic)
6. **Undesirable** (Pacing, stereotypies, self-injury, etc.) — include only if you see it
7. **Not Visible** (Behavior Obscured + Animal Not Visible)

Your required categories: choose **exactly 5**, and one MUST be **Not Visible**. Write your 5 categories here:

1. ______________________________

2. ________________________________

3. ________________________________

4. ________________________________

5. ________________________________

Tip: If you never see "Undesirable" or "Social," don't include them—pick categories you expect to observe at least sometimes.

Step C — Collect data using scan sampling

You will do **20 scans total** (10 scans per condition).

- Set a timer for **1-minute intervals**
- At each minute mark, record the **single best behavior category** that describes what the animal is doing **at that instant**
- If more than one behavior occurs, score the behavior that is **most dominant** at the instant of the scan
- If you cannot determine behavior, score **Not Visible** (Obscured or Not Visible)

Scan sheet

Fill in one row per scan.

Scan #	Condition (A or B)	Behavior category (one of your 5)	Notes (optional)
1	A		
2	A		
3	A		
4	A		
5	A		
6	A		
7	A		
8	A		
9	A		
10	A		
11	B		
12	B		
13	B		
14	B		
15	B		
16	B		
17	B		
18	B		
19	B		
20	B		

Step D — Convert your scans into a summary count table

Count how many times each category occurred in each condition.

Observed counts (O)

Behavior category (your 5)	Condition A (count)	Condition B (count)	Row total
Category 1			
Category 2			
Category 3			
Category 4			
Category 5			
Column totals	**10**	**10**	**20**

(If your totals aren't 10 and 10, you missed scans—fix before continuing.)

Part 3: Hypotheses (chi-square test of independence)

Your research question (one sentence)

Example formats:

- "Does the distribution of behaviors differ between Condition A and Condition B?"
- "Does Animal A show a different behavior profile than Animal B?"

Question:

Null hypothesis (H0)

For a chi-square test of independence:

H0: The proportion of scans in each behavior category is the **same** in Condition A and Condition B.

Write it in your own words:

____

Alternative hypothesis (HA)

HA: The distribution of behaviors is **different** between Condition A and Condition B.

Write it in your own words:

____

____

Step E — Calculate expected counts (E)

For each cell:

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

Because each column total is 10 and grand total is 20:

$$E = \frac{(\text{row total})(10)}{20} = \frac{\text{row total}}{2}$$

So for each behavior category, the expected count in Condition A equals **half of the row total**, and same for Condition B.

Expected counts (E)

Behavior category	Row total	Expected in A (E)	Expected in B (E)
Category 1			
Category 2			
Category 3			
Category 4			
Category 5			

Step F — Compute chi-square

For each cell:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Chi-square work table

Fill in all 10 cells (5 categories × 2 conditions).

Behavior category	O (A)	E (A)	(O−E)	(O−E)²/E	O (B)	E (B)	(O−E)	(O−E)²/E
Category 1								
Category 2								
Category 3								
Category 4								
Category 5								
Totals				**chi² =**				

Step G — Degrees of freedom and decision

Degrees of freedom for chi-square test of independence:

$$df = (r - 1)(c - 1)$$

Here, (r=5) behavior categories and (c=2) conditions, so:

$$df = (5 - 1)(2 - 1) = 4$$

- Use your course method (table or calculator/software) to find the **p-value** for your ^{2} with df = 4.
- Use (= 0.05) unless your instructor tells you otherwise.

chi-square value: ______________________________

df: 4

p-value: ______________________________

Decision:

- [] Reject H0
- [] Fail to reject H0

Interpretation (2–4 sentences)

- What does your result mean in plain language?
- Which behaviors contributed most to ^{2} (largest $(O-E)^2/E$ values)?

Data quality checks (answer briefly)

1. Did you have any categories with expected counts **less than 5**?

- [] No
- [] Yes
  - If yes, combine two similar categories and re-run the test.

2. Were there many **Not Visible** scans?

- [] No
- [] Yes
  - Briefly explain why (camera angle, animal location, exhibit complexity, etc.).

3. One limitation of your observational design:

Submission checklist

Turn in:

- Completed scan sheet
- Observed count table (O)
- Expected count table (E)
- Chi-square work table with ^{2}, df, p-value
- Short interpretation + data quality checks